

Technical Documentation

Fatigue-Guard: Real-time EMG-Based Muscle Fatigue Detection Wearable

1) System Overview

Fatigue-Guard is a portable wearable electromyography sensor designed to detect muscle fatigue in real-time during strength training and athletic activities. The system monitors electrical activity within working muscles and alerts users before form degradation and injury risk escalate. Our approach uses time-domain amplitude tracking to establish personalized baselines for each training set, enabling early fatigue detection without requiring clinical-grade equipment or expensive laboratory setups.

The implemented system consists of three primary components. A body-worn wearable captures electromyography signals directly from the skin surface. Local signal conditioning hardware amplifies and filters raw muscle signals. Embedded firmware interprets the conditioned signal, compares it against a fresh baseline established at the start of each set, and triggers haptic and visual alerts when muscle fatigue reaches a threshold. The user interface combines vibration feedback and an OLED display to communicate fatigue status without requiring the user to check a separate device.

1.1 System Architecture

- Body interface: Dry electrode configuration on the forearm or upper arm captures muscle electrical potentials
- Signal acquisition: Instrumentation amplifier stage multiplies weak EMG signals by 2,420x and prepares them for downstream filtering
- Analog processing: Bandpass filtering removes heartbeat interference and high-frequency electrical noise, isolating the muscle contraction band
- Digital conversion: An ADC on the microcontroller digitizes the conditioned signal at 1 kHz sampling rate
- Firmware processing: Running moving-average calculations and threshold comparisons in real-time to detect fatigue onset
- User alerts: A vibration motor provides tactile feedback and an OLED display flashes a REST NOW message when fatigue threshold is reached

2) Implemented Functionality

| Function                    | Current Implementation                                                                                                                                                                 |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Electrode attachment        | Dry electrode pads placed on muscle belly. User cleans skin and applies electrodes to forearm or arm depending on exercise focus.                                                      |
| Set baseline capture        | A phase check gate identifies the start of muscular contraction. Moving average of the first few reps is stored as the fresh baseline for that set.                                    |
| Real-time fatigue threshold | Firmware continuously updates a moving-average window of the signal envelope. When amplitude increases beyond a learned threshold relative to baseline, the fatigue flag is triggered. |
| Haptic and visual alert     | Wrist-mounted vibration motor activates repeatedly. OLED display on the device shows REST NOW message in high contrast.                                                                |

## 3) Hardware Specification

### 3.1 Materials

| No. | Component                              | Qty       | Notes                    |
|-----|----------------------------------------|-----------|--------------------------|
| 1   | ESP32 DevKit V1                        | 1         | Main processing unit     |
| 2   | Dry EMG electrode pads                 | Multiple  | Replaceable consumable   |
| 3   | Instrumentation amplifier (INA128)     | 1         | Signal amplification     |
| 4   | Bandpass filter module (72-720 Hz)     | 1         | Analog filtering         |
| 5   | 0.96 inch OLED display                 | 1         | User visual alert        |
| 6   | Coin vibration motor                   | 1         | Haptic alert feedback    |
| 7   | N-Channel transistor (2N7000)          | 1         | Motor control switch     |
| 8   | 3.7 V LiPo battery with TP4056 charger | 1         | Power supply             |
| 9   | Breadboard and jumper wires            | As needed | Assembly and prototyping |

### 3.2 Construction Notes

- The prototype is built on breadboard with minimal hand-soldered connections. No custom PCB has been manufactured at this stage.
- Housing is currently a wrist strap with elastic material. Future iterations would use injection-molded plastic or 3D printed enclosures.
- Electrode contact is dry paste applied directly to skin. The design avoids gel-based electrodes to reduce cost and consumable dependency.
- Signal conditioning is entirely analog. No digital filtering or adaptive algorithms run on the conditioning stage itself.

### 3.3 Microcontroller Pin Map

| ESP32 Pin           | Connected To             | Function                                            |
|---------------------|--------------------------|-----------------------------------------------------|
| GPIO 34             | INA128 output            | Analog-to-digital conversion of EMG signal at 1 kHz |
| GPIO 25             | Transistor gate          | Controls vibration motor via transistor switch      |
| GPIO 21 and GPIO 22 | OLED display SDA and SCL | I2C communication with display module               |
| 3.3 V and GND       | All analog circuits      | Power and common ground distribution                |

## **4) Firmware and Signal Processing**

### **4.1 Signal Acquisition and Baseline Establishment**

The firmware initializes the ADC on GPIO 34 to sample at 1 kHz. Upon power-up or set start, the device enters a baseline acquisition phase. During this phase, it calculates a moving average of the incoming EMG signal amplitude over the first five to ten muscle contractions. This moving average serves as the fresh baseline for that specific set. The baseline captures the resting or lightly activated state of the muscle.

The phase check gate identifies when the user begins a lift. It monitors for a threshold crossing that indicates the start of active contraction. Once detected, the firmware stores samples and computes statistics. The gate prevents false baseline captures caused by minor tremor or noise.

### **4.2 Real-Time Fatigue Detection**

During active repetitions, the firmware continuously updates a time-domain moving average of the signal envelope. It compares this envelope against the established baseline. When the envelope amplitude rises above a programmed threshold relative to baseline, the fatigue flag is set to true. The threshold is calibrated during prototype testing to trigger near the point when visible form degradation becomes likely.

The detection is conservative to minimize false positives. It does not rely on frequency domain features or machine learning. Instead, it uses straightforward moving average comparisons that can run efficiently on embedded systems without external processors.

### **4.3 User Alerting and Reset**

When the fatigue threshold is crossed, the firmware immediately triggers the vibration motor via a GPIO output that drives a transistor. The transistor gate on GPIO 25 receives a high signal, causing the motor to pulse repeatedly. Simultaneously, the OLED display shows a REST NOW message with high contrast. The alert pattern repeats every 300 milliseconds until the user acknowledges the fatigue state by resting or powering down the device.

At the end of the set, the user can trigger a reset by pressing a capacitive touch button or by waiting for the next baseline capture cycle to begin. The device is then ready to establish a fresh baseline for the next set.

## **5) Performance Metrics and Validation**

Field testing of the prototype was conducted under gym conditions during free weight exercises. A test subject performed bicep curls while wearing the wearable on the forearm. The device successfully detected fatigue onset when muscle output visibly declined and form began to deteriorate. The baseline captured during the initial repetitions remained stable across all subsequent reps until the threshold was crossed in the final two or three reps of the set.

Signal quality was adequate with the dry electrode configuration and minimal electrical noise from the lab environment. The bandpass filter effectively removed low frequency heartbeat interference and high frequency power line hum. No special skin preparation beyond basic cleansing was required for reliable operation.

Battery life under continuous operation was estimated at four to six hours based on prototype current draw measurements. Future iterations with power optimization could extend this significantly. The cost of the complete system in prototype form is approximately one tenth the cost of commercial EMG systems.

## **6) Technical Limitations and Future Work**

The current prototype has several limitations that should be noted. The time-domain approach does not capture frequency-based fatigue signatures that may provide additional predictive power. Electrode placement is manual and user-dependent, which could introduce variability in signal quality. The device does not store historical data or sync to a companion mobile application. Integration with external devices or cloud platforms has not been implemented. Controlled testing in diverse gym settings and across different exercises and muscle groups has not yet been completed.

Future development should explore custom PCB design to reduce component count and power consumption. Adaptive machine learning algorithms trained on larger datasets could improve accuracy across broader populations. A mobile companion app would enable data visualization and personalized threshold tuning. Testing with professional athletes and coaches would validate the practical utility of the system in real training scenarios.

## **7) Conclusion**

Fatigue-Guard demonstrates that real-time muscle fatigue detection is achievable with affordable wearable hardware and straightforward digital signal processing. The prototype successfully captures EMG signals, establishes personalized baselines, and alerts users when fatigue approaches critical levels. By bridging the gap between expensive clinical systems and consumer fitness trackers, this approach offers a practical solution for injury prevention in strength training and athletic contexts.